

Developer Salary Prediction

ML1 Task 2 — Regression

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Problem

Goal: Predict annual developer compensation (`annual.pay.usd`)

Metric: Root Mean Square Error (RMSE)

Constraint: ML1-syllabus models only (no trees)

Training rows	~2 500
Test rows	628
Target	Annual salary in USD (heavy-tailed)
Key challenge	Multi-select survey columns, extreme outliers

Three decisions made before any model:

1. Multi-select columns (tech stack, tools, languages) — multi-hot + count features
2. Target modelled in **log-space** to stabilise variance
3. Salary outliers handled by **range filter** [1k, 300k]

Feature Engineering

Multi-select columns (technologies, tools, languages, frameworks):

- ▶ Multi-hot encode each column (`min_freq = 100`, `top_k = 10` per column)
- ▶ Add **count feature** per column — breadth-of-experience proxy
- ▶ Derived: `professional_experience_share`, `non_professional_years`

Target encoding (8 high-cardinality columns):

- ▶ `region`, `industry`, `company.size`, `age.group`, `dev.role`, `education`, `employment.type`, `work.location`
- ▶ Smoothed K-fold target encoding — no leakage across folds

Preprocessing:

- ▶ `StandardScaler` + mean imputation
- ▶ Salary filter [`1 000`, `300 000`] applied only to training data
- ▶ All steps fitted on training fold only

Algorithms Considered

5-fold cross-validation on the same pipeline, RMSE in USD:

Model	Unfiltered data	After range filter
Ridge	93 363	39 482
Lasso	90 980	38 232
Elastic Net	91 848	38 702
KNN	87 898	38 826
SVR (RBF)	86 295	36 659
Random Forest	84 277	34 383
Gradient Boosting (<i>reference</i>)	84 194	34 658

- ▶ Range filtering alone cuts RMSE by ~55 k — extreme salaries dominate squared error
- ▶ **SVR is beaten by RF - but in the end it was our best final model**
- ▶ Linear models lag significantly — salary signal is non-linear

Hyperparameter Tuning

Three rounds of tuning for SVR:

1. **Preprocessing sweep** — scaler, imputer, min_freq / top_k for multi-hot encoding
2. **SVR grid search** — C in {1, 2, 3, 5}, epsilon in {0.03, 0.05, 0.08}, gamma in {scale, auto, 0.005, 0.01}
3. **Target encoding ablation** — progressively adding high-cardinality columns

Tuning stage	5-fold CV RMSE
Initial SVR (default params)	36 659
After preprocessing tuning	33 395
+ SVR grid search	31 488
+ Target encoding (8 cols)	31 284

Selected: SVR(kernel=rbf, C=1, eps=0.03, gamma=auto) with StandardScaler and mean imputation

Ensembling

Out-of-fold (OOF) blend of SVR + Random Forest + Ridge:

- ▶ Each model produces OOF predictions on the training set
- ▶ Weights optimised with Nelder-Mead on OOF RMSE
- ▶ Best found blend: **SVR 90.3% + RF 9.7%** (Ridge weight collapsed to ~ 0)

Configuration	OOF RMSE (eval on [1k, 300k])
SVR alone	31 284
SVR + RF + Ridge (Nelder-Mead)	30 984

Training on full data:

- ▶ Blend weights fixed from OOF optimisation
- ▶ Both SVR and RF retrained on all $\sim 2\,300$ filtered rows
- ▶ Test predictions: 90.3% SVR + 9.7% RF (on log-scale, then exponentiated)

Why the Blend

Argument	Detail
OOF gain is real RF complements SVR	-300 RMSE on 5 folds, consistent across splits RF catches mid-range clusters SVR smooths over
Tiny RF weight	9.7% limits risk of overfitting RF to training noise
Ridge dropped	Ridge too weak (38k) to add value alongside SVR (31k)

Risks acknowledged:

- ▶ OOF evaluation uses filtered salaries [1k, 300k]; true test set may include extreme values
- ▶ Expected ~2–3 k CV-to-LB gap: range filter before CV split makes CV optimistic

Bottom line: SVR alone at ~31 k is the “safe” submission; the blend at ~31 k is a ~300-point improvement with limited downside.

Expected Results

Final pipeline (end-to-end):

1. Salary filter [1k, 300k] on training data only
2. Multi-hot encode + count / derived features
3. K-fold target encode 8 high-cardinality columns
4. StandardScaler + mean imputation
5. **SVR** (RBF, C=1, eps=0.03, gamma=auto) fit on log-salary
6. Optional: blend 90.3% SVR + 9.7% RF predictions
7. `exp(prediction)` → submission

Estimate	Value
5-fold CV RMSE (filtered eval, SVR alone)	31 284
5-fold CV RMSE (OOF blend)	30 984
Expected leaderboard RMSE	~33–34 k
CV-to-LB gap	~2–3 k (structural, range filter artefact)

Summary

Best model	SVR with RBF kernel
Why SVR	Best ML1-allowed model; non-linear; robust to feature scale
Biggest win	Range filtering — removed extreme salaries (-55 k RMSE)
Second win	Smoothed K-fold target encoding (-5 k RMSE)
Blend gain	SVR 90.3% + RF 9.7% (-300 RMSE over SVR alone)
Expected LB	~33–34 k RMSE

Tried and rejected:

- ▶ Winsorization — hurt CV by ~4 k (clipped targets in validation folds)
- ▶ SVR + Ridge blend — Ridge too weak; blend pulled SVR down
- ▶ Stacking (meta-learner) — did not outperform the tuned single SVR

Near the data-driven ceiling for ML1-syllabus models on this dataset.